



Analog MEMS Electronic Stethoscope

Simulation & PCB Designs

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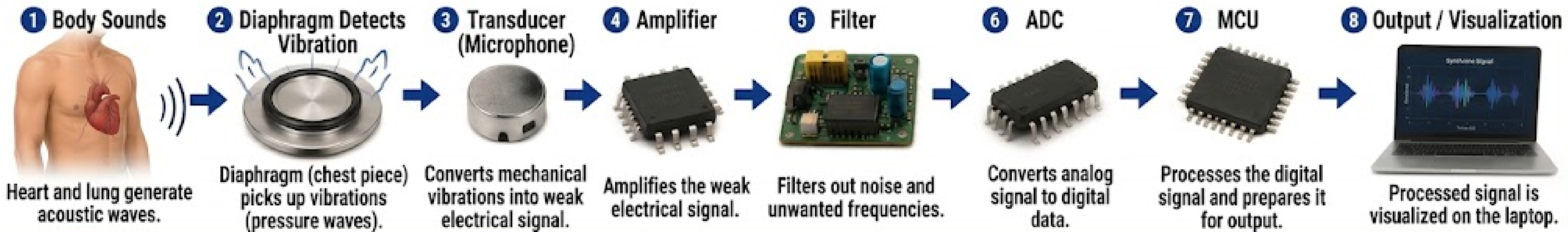


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- Review of Existing Analog Front-End (AFE) Designs
- Our Proposed Analog Front-End Design
- Remaining System Design and Power Management



System Overview





Analog Front End Design I (Non inverting amplifier)

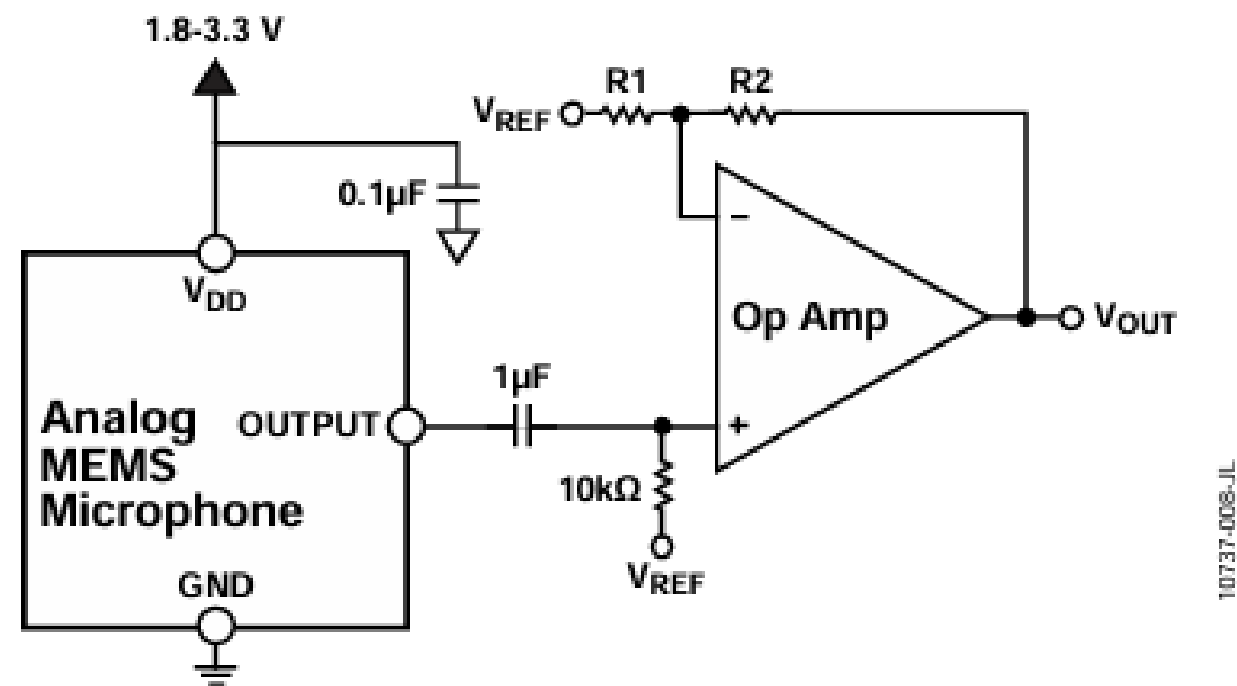


Figure 6. Analog Microphone Connection to Noninverting Op Amp Circuit

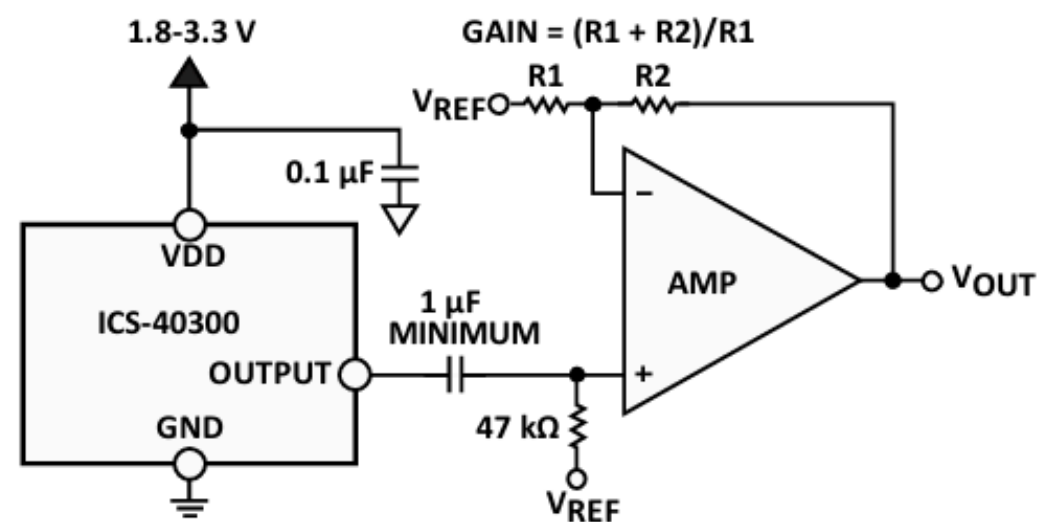
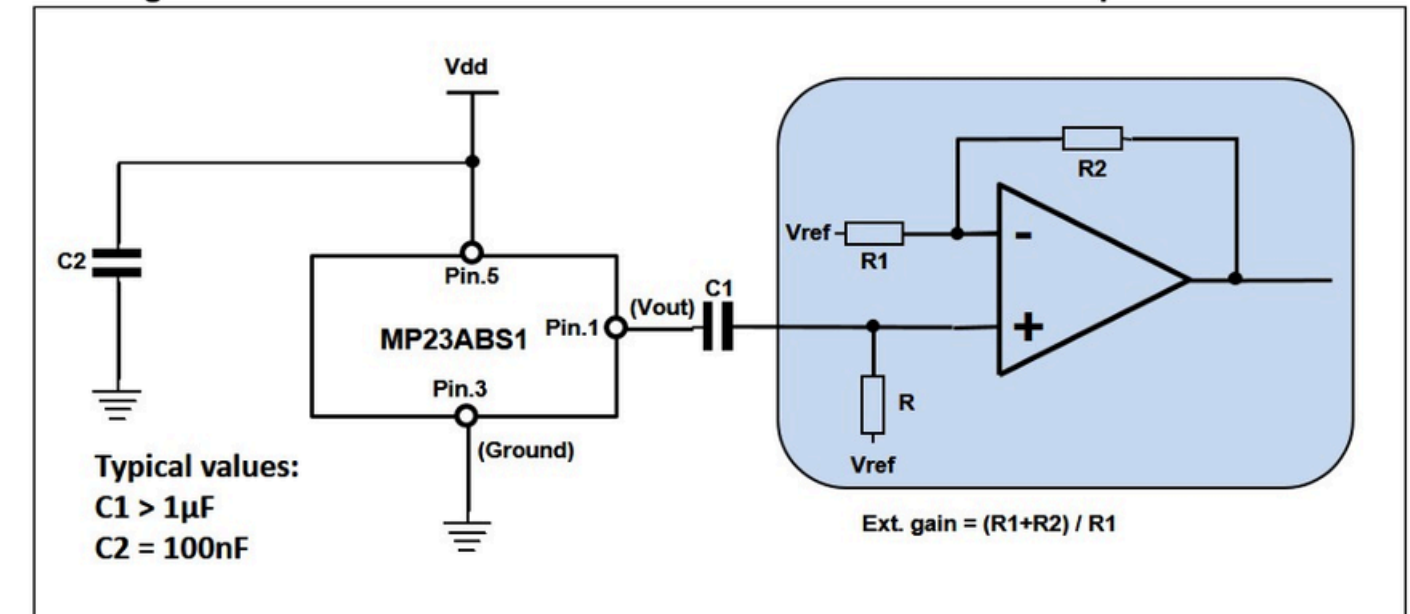


Figure 13. ICS-40300 Connected to an Op Amp

MP23ABS1 schematic hints

Figure 3. MP23ABS1 electrical connections and external component values



- high input impedance
- prevent the loading of the microphone



Analog front end Design III (Interting Amplifier + Anti aliasing filter)

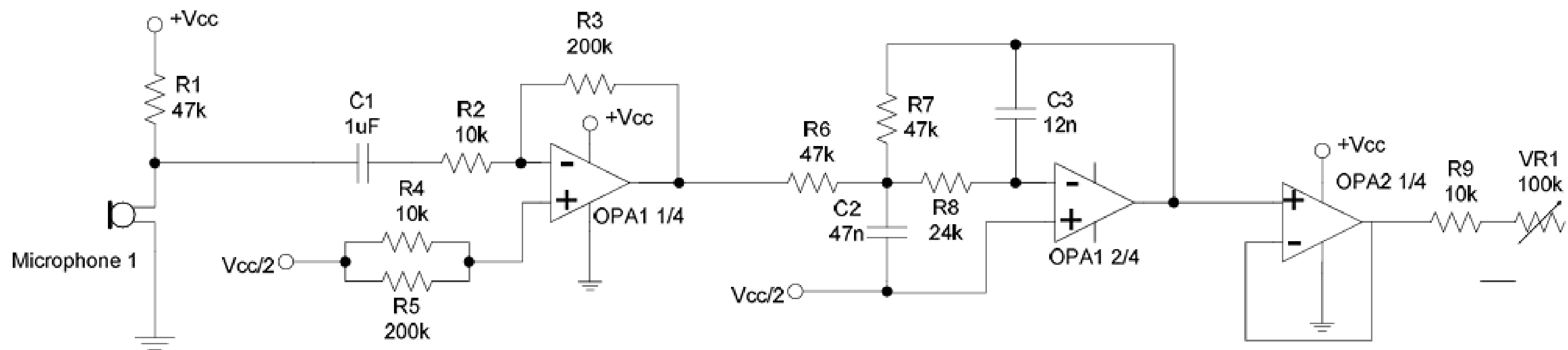


Fig. 6. Pre-amplifier and the anti-aliasing circuit.

- Preamplifier with gain of 20
- anti aliasing low pass filter cutoff 2kHz



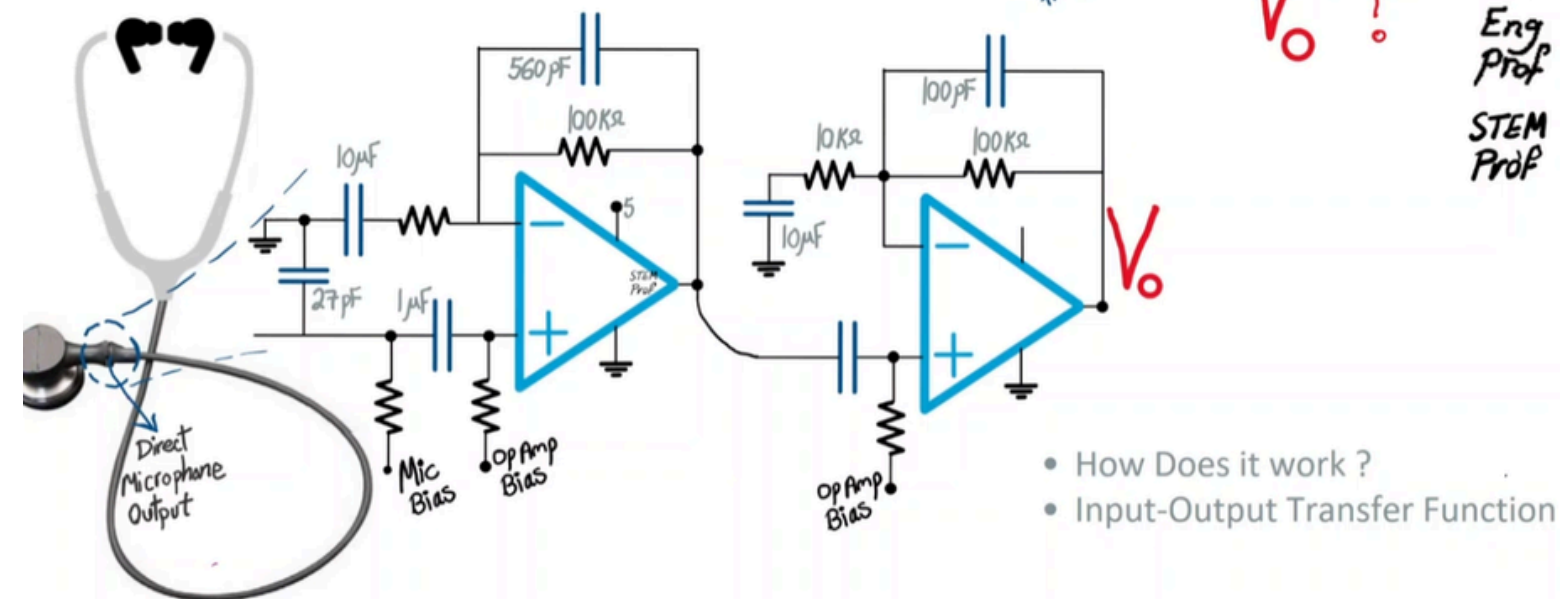
Analog front end Design IV (active Band pass filter)

Digital Stethoscope Amplifier

Video #214

V_o ?

© Eng Prof STEM Prof



- How Does it work ?
- Input-Output Transfer Function

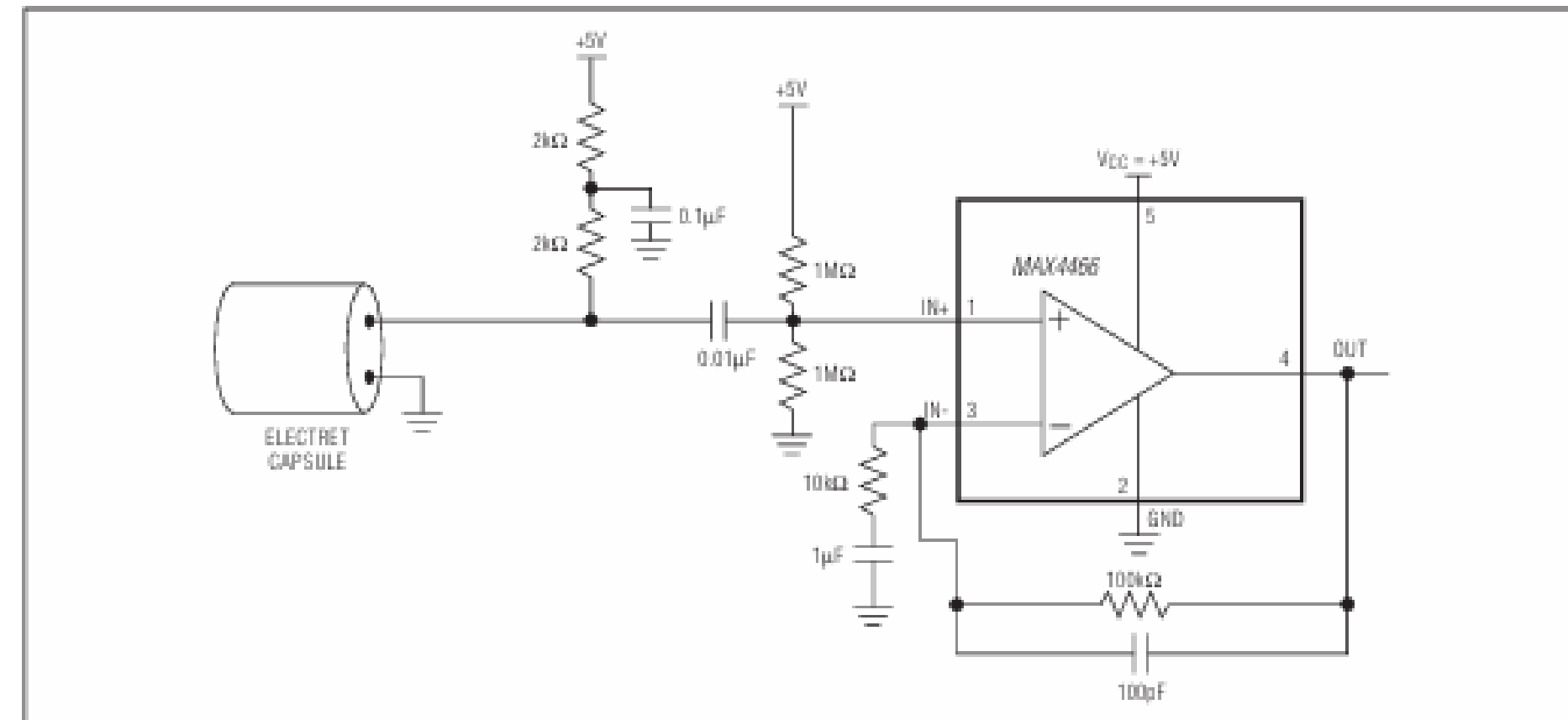


Figure 2. MAX4466 Typical Application Circuit

- 2 stage band pass filter
- cut off frequency 3.5Hz ,150Hz



Analog front end Design V (Two stage amplification)

The Complete Working Circuit Diagram.

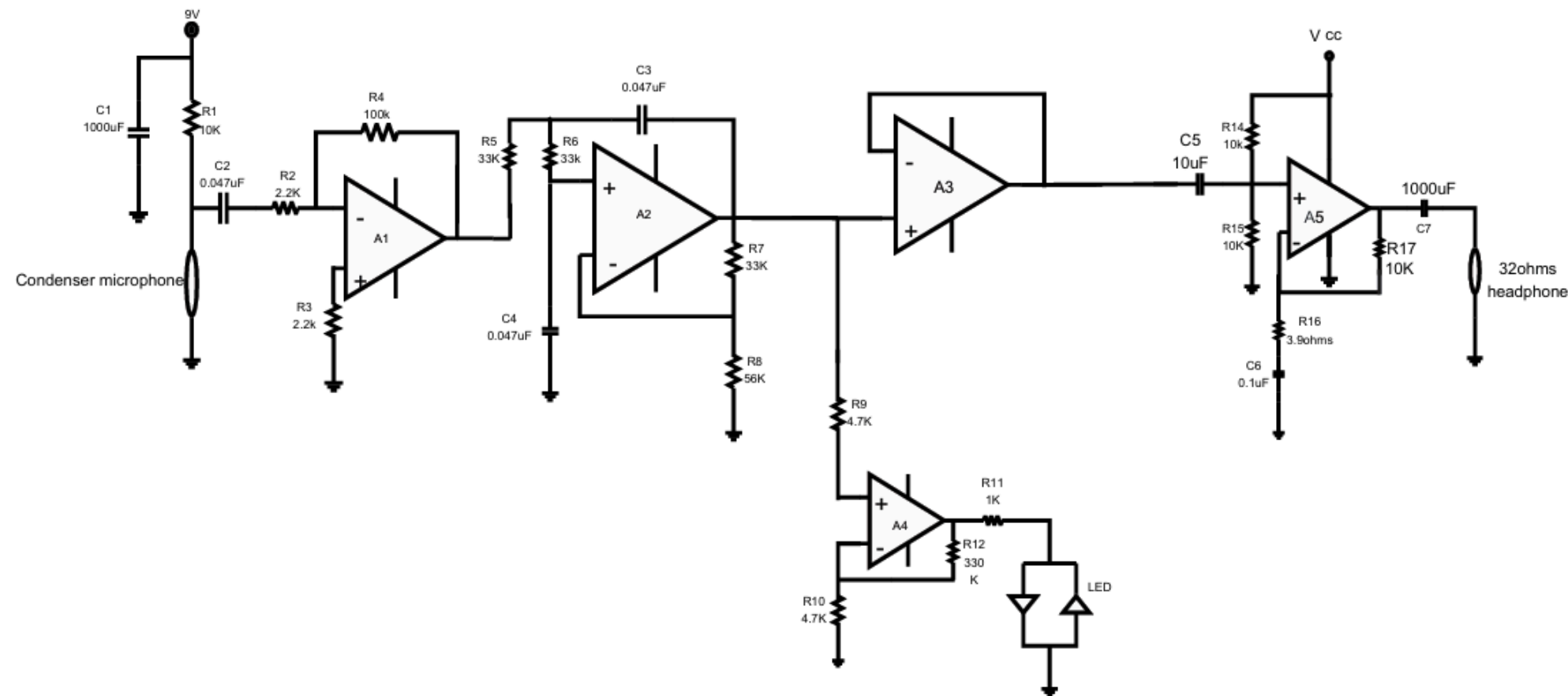


Figure 2.8 Schematics of the Electronic Stethoscope

- Dual polarity power source
- pre amplifier
gain - 45
closed loop bandwidth - 22kHz
- Second order butterworth low pass filter
cut off of 100Hz , gain 1.6
- Voltage follower



Literature-Based System Block Diagrams

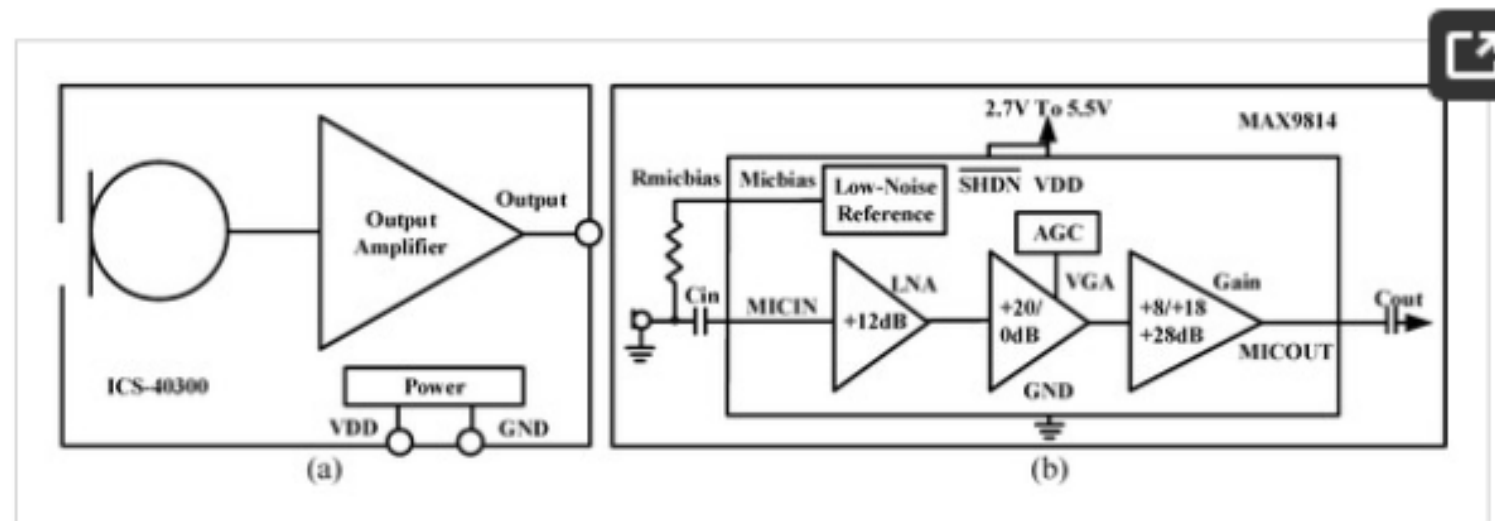
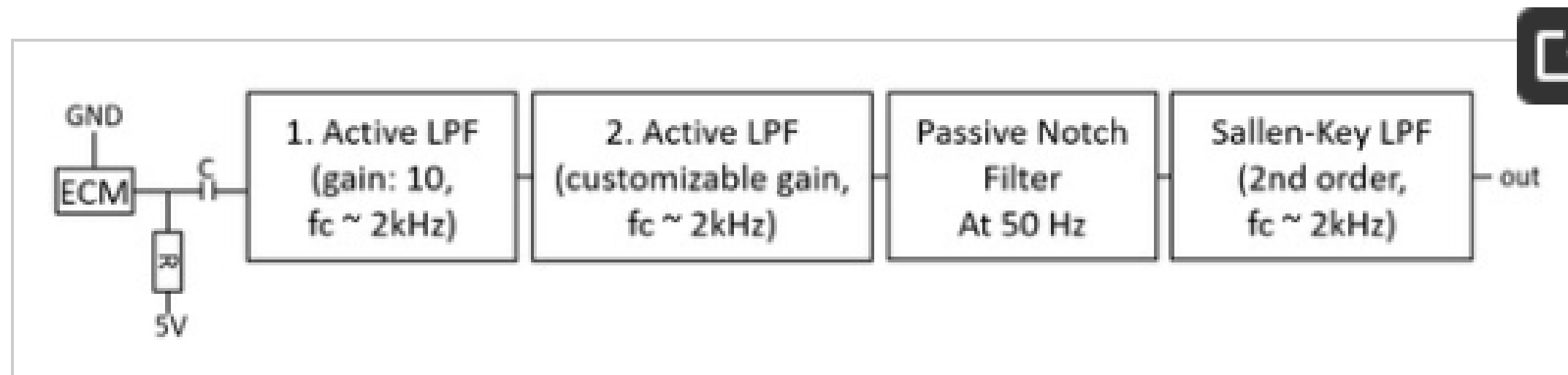


Figure 2. Schematics of (a) ICS-40300 and (b) MAX9814.



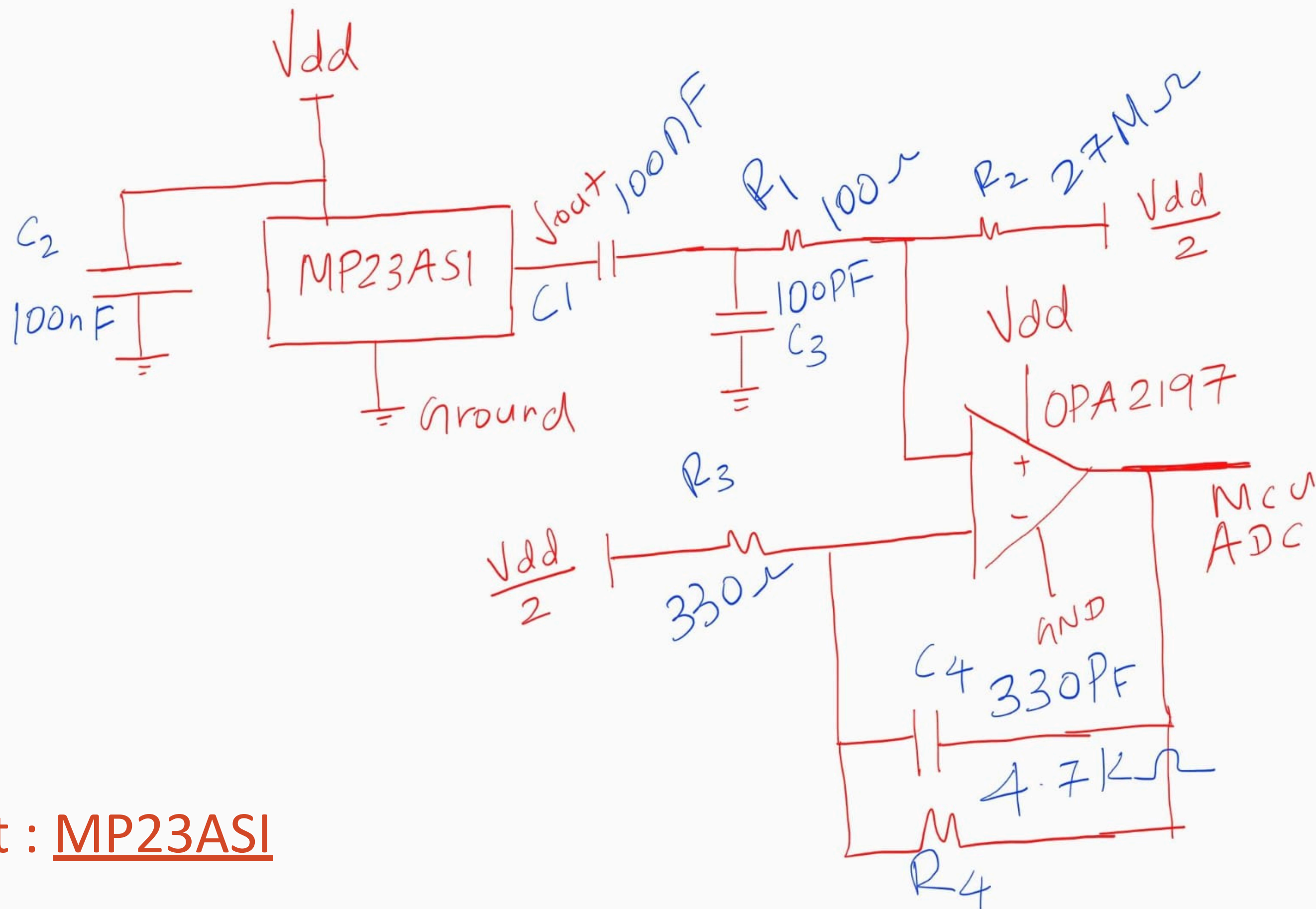
Source -

Development of a Multi-Channel Wearable Heart Sound Visualization System By Binbin Guo,Hong Tang, hufeng Xia, Miao Wan, Yating Hu and Zehang Zhao

Source - <https://doi.org/10.3390/app152111569>



Circuit Design 01



Datasheet : [MP23AS1](#)



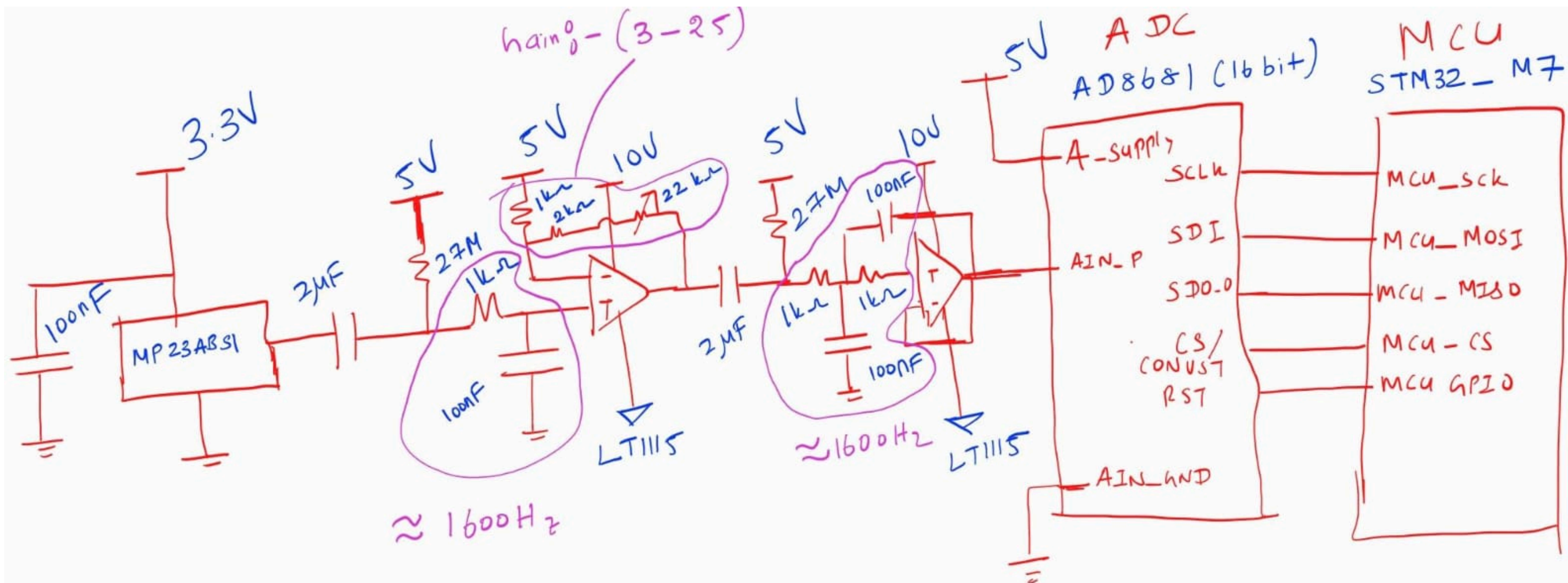
Final Circuit with Specification

MEMS	Sensitivity	Output Type
<u>MP23ABS1</u>	-38 dBV	Single end
<u>MP23AB01DH</u>	-38 dBV	Differential
<u>ICS-40300</u>	-45 dB V	Single end

OPAMP	Noise input	Application
<u>LT1115</u>	Very low	High Quality Audio Preamplifiers
<u>LM833-N</u>	Compared to LT1115 High	Audio System
<u>OPA2197</u>	Compared to LT1115 High	Battery Management



Final Circuit with Specification





Final Circuit with Specification

ADC

$$\begin{aligned} &= \frac{V_{ref}}{2^N} \\ &= \frac{4.096}{2^{16}} \\ &= 62.5 \mu V // \end{aligned}$$

Selected Sampling Rate of ADC
0 - 4 kHz (from lit)
By using Eng. Safety Margine 0 - 2.5
Nyquist frequency = $\frac{4000}{2.5} = 1600 \text{ Hz}$

Filter

Anti Aliasing filter (low pass) 0 - 1600 Hz (cutoff)

$$\left. \begin{aligned} &\frac{1}{2\pi RC} = 1600 \\ &R = 1 \text{ kHz} \\ &C = 100 \text{ nF} \\ &f_c \sim 1.59 \text{ kHz} \end{aligned} \right\}$$

MEMS

SPL (dB SPL)	Sensitivity (dB V)	around frequency (Hz)
92	-38	1 kHz
80	-50	1 kHz
40	-90	1 kHz
92	-40	20 Hz
80	-52	20 Hz
40	-92	20 Hz

$$\text{Voltage level } 0 - 10^{\left(\frac{\text{dB SPL}}{20}\right)}$$

$$\begin{aligned} 25.1 \mu V & \text{ --- } 31.6 \mu V & \text{ --- } 40 \text{ dB SPL} \\ 2.51 \text{ mV} & \text{ --- } 3.16 \text{ mV} & \text{ --- } 80 \text{ dB SPL} \\ 0.01 \text{ V} & \text{ --- } 0.012 \text{ V} & \text{ --- } 92 \text{ dB SPL} \end{aligned}$$

According to lit survey (40 dB - 80 dB) SPL
Restricted to that

$$\begin{aligned} \text{Min gain} &= \frac{\text{ADC Unit step}}{\text{Min Sensitive voltage}} \\ &= \frac{62.5 \mu V}{25.1 \mu V} \\ &= 2.49 // \end{aligned}$$

Filter order

We selected third order filter for sharp roll off (-60 dB) — As optimal for 80 dB

$$\begin{aligned} \frac{V_{max}}{1000} &\approx 62.5 \mu V \\ V_{max} &\approx 62.5 \text{ mV} \end{aligned}$$

$$\begin{aligned} V_{max} \text{ before preamp} \times \text{Max gain} &= 62.5 \text{ mV} \\ V_{b \text{ preamps}} &= \frac{62.5 \text{ mV}}{2.51 \text{ mV}} \\ &= 24.9 \\ &\approx 25 // \end{aligned}$$

But this will saturate the ADC @ 92 dB SPL not fully used @ 40 dB SPL
Using lit. Analysis Max used is 20
so we selected changeable gain

$$0 - (3 - 25)$$

$$1 + \frac{R_1 + R_{pot}}{R_{rev}} = (3 - 25)$$

$$\frac{R_1 + R_{pot}}{R_{rev}} = (2 - 24)$$

$$\begin{aligned} R_1 &= 2 \text{ k}\Omega // \\ R_{pot} (\text{Max}) &= 22 \text{ k}\Omega // \\ R_{rev} &= 1 \text{ k}\Omega // \end{aligned}$$



Microcontroller Selection

MCU COMPARISON Scoring: 1 = Poor 5 = Excellent

Requirement	STM32H755BI	ESP32	STM32L476RG
DSP Capability	5	2	4
Battery Life	1	4	5
Architecture Fit	3	2	5
Development & Predictability	1	3	4
TOTAL	10/20	11/20	18/20

FINAL SELECTION

- STM32L476RG
- CPU: ARM Cortex-M4
- Flash: 1 MB
- SRAM: 128 KB
- DSP: Hardware DSP instructions
- Real-Time Processing: DMA + NVIC
- Run-Mode Current: ≈ 3.0 mA



- Using the dedicated RNBD451PE for Bluetooth allows the STM32L476RG to focus on stable, real-time DSP processing with low power consumption of 3.0 mA.



Wireless Communication Strategy

BLUETOOTH vs. WI-FI

WHY BLUETOOTH?

- Lower power consumption
- Direct point-to-point connection
- Suitable for battery-powered operation

PCG DATA RATE CALCULATION

$$\begin{aligned}\text{Data Rate} &= \text{Sampling Rate} \times \text{Resolution} \times \text{Channels} \\ &= 4000 \text{ Hz} \times 16 \text{ bits/sample} \times 1 \\ &= 64,000 \text{ bps} \\ &= 64 \text{ kbps}\end{aligned}$$

BANDWIDTH SUITABILITY

- RNBD451PE supports up to 2 Mbps
- 64 kbps required $\ll$ 2 Mbps available

SELECTED MODULE - Microchip RNBD451PE

- Bluetooth: Bluetooth 5.2 Low Energy
- Supply Voltage: 1.9 V – 3.6 V
- Typical: 3.3 V
- TX Power: Up to +15 dBm
- RX Sensitivity: -95.5 dBm & 1 Mbps





System Power Estimation & Battery selection

Subsystem Current Breakdown

Subsystem	Components	Estimated Current
Analog Front-End (AFE)	MEMS Microphone	3.5 mA
Microcontroller (MCU)	STM32L476RG	8.5 mA
Bluetooth LE Module	Microchip RNBD451PE	15.0 mA
System Current (I _{avg})	All Subsystems	27.0 mA

FINAL BATTERY SELECTION

- Battery: Single-Cell Li-Po
- Nominal Voltage: 3.7 V
- Capacity: 1000 mAh
- Energy: 3.7 Wh
- Operating Range: 4.2 V → 3.0 V



BATTERY CAPACITY CALCULATION

1. THEORETICAL CAPACITY

$$C_{\text{theoretical}} = I_{\text{avg}} \times T_{\text{required}}$$

$$C_{\text{theoretical}} = 27 \text{ mA} \times 24 \text{ h}$$

$$C_{\text{theoretical}} = 648 \text{ mAh}$$

PRACTICAL CAPACITY

$$\text{Safety factor} = 1.25$$

$$C_{\text{required}} = C_{\text{theoretical}} \times 1.25$$

$$C_{\text{required}} = 648 \times 1.25$$

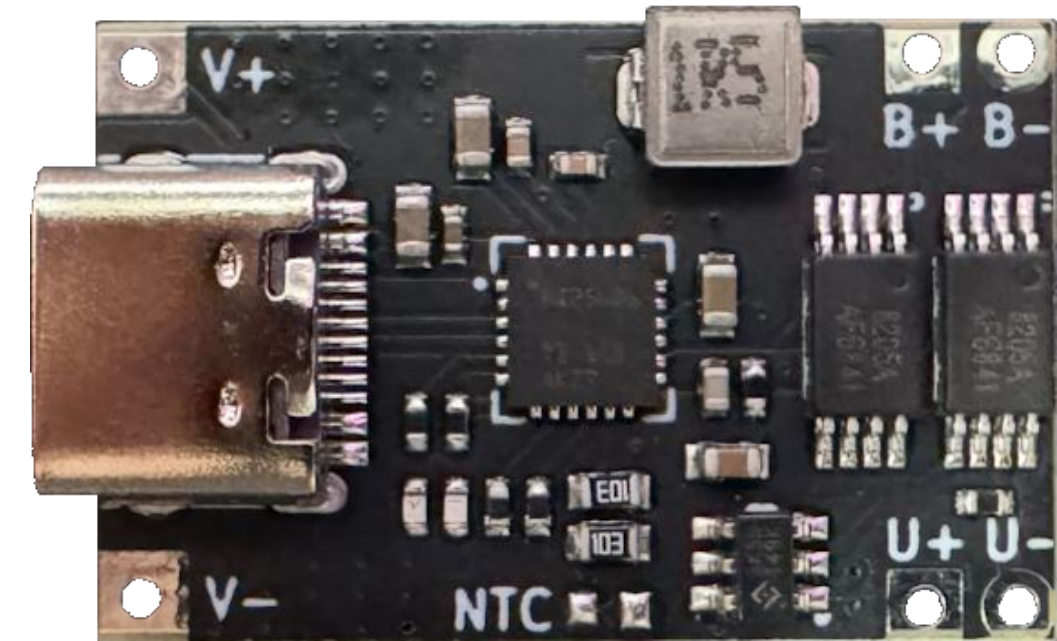
$$C_{\text{required}} = 810 \text{ mAh}$$



Selected Charging IC

SELECTED CHARGING IC - BQ25606

- Input Voltage - 3.9 V – 13.5 V
- Battery Regulation Voltage - 4.208 V / 4.352 V / 4.400 V
- Maximum Charging Current - Up to 3.0 A



SUITABILITY

Voltage Compatibility

- 4.208 V regulation matches the maximum charging voltage of the selected 3.7 V Li-Po battery

Input Flexibility

- 3.9 V – 13.5 V input supports standard 5 V USB charging.

SAFETY

Charge Termination

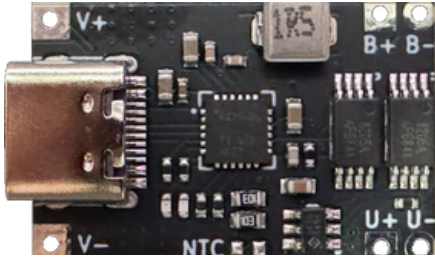
- Automatically stops charging when the battery current falls below the termination threshold.

Thermal Regulation

- Charging current is automatically reduced when junction temperature exceeds 110 °C.



Final Component list

Component	Model	Image	Datasheet
Micro Controller	STM32L476RG		https://www.alldatasheet.com/datasheet-pdf/view/929993/STMICROELECTRONICS/STM32L476RG.html
Charging IC	BQ25606		https://www.alldatasheet.com/html-pdf/935960/TI1/BQ25606/54/1/BQ25606.html
Recharchable battry	Single cell Li - Po -1000mAh		https://www.fpbattery.com/wp-content/uploads/2024/06/fpbattery-951213-3.7V-150mAh-Lithium-Polymer-Battery-Specification.pdf
Bluetooth Module	RNBD451PE		https://www.microchip.com/en-us/product/rnbd451pe



Progress Review

THANK YOU